in series. This MCPCB is a round PCB used in 5W LED bulbs. Mount it on an aluminium channel.

Resistor value calculations

1. Voltage at maximum power Vpv = 17.5 × 7 = 122.5V

2. Number of LEDs in each lamp = 24 + 7 + 5 = 36

3. Total forward voltage of LEDs = 36 × 3 = 108V

4. Voltage difference = 122.5 - 108 = 14.5V

5. LED forward current = 0.25A (maximum allowed 0.33A)

6. Value of series resistance required = 14.5/0.25 = 58Ω

7. Power dissipated in resistor = 0.25 × 0.25 × 58 = 3.5W

Standard resistor value selected for R1, R2, and R3 is 56Ω, 5W. Photographs of the SDL are shown in Fig. 6 and Fig. 7.

This lamp provides light throughout the day. It can be installed at homes, apartments, offices, small shops, storage area of kirana shops, etc. Wherever there is darkness during daytime, all those places can get good amount of light thoughout the day. It is necessary to have lamp(s) working on mains supply as a backup. Whenever the light from SDL is insufficient, the user can turn on the mains lamp(s).

Fig. 7: Close up of the round MCPCBs with 7 and 5 LEDs

Important note. PV system described here is just a proposal. For performance evaluation, we need to implement it in a few apartments. Then, based on the outcome of the evaluation, it can be installed in many more places.

This PV system for apartments is proposed in spite of so many limitations. Hence, we may not get very high returns. Therefore, it is not for those who are looking for big savings in their power bills. Rather, this is for those who want to contribute to the cause of environment. It will take some time before the system matures and many more applications are identified. Along with that, the improvement in solar cell efficiency will help in getting more output. With that, we can expect good returns from the system.

Meanwhile, we can derive other benefits of the off-grid system such as higher reliability and uninterrupted service for many years. This is useful for remote locations where the grid supply is not available. EFY

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